

KonthoProhori Model And Data Card

One-page AI component summary for SciBlitz AI Challenge 2026

Dataset	Details
Name and source	Bangla DeepFake Dataset by ahanaf101 on Kaggle
URL	https://www.kaggle.com/datasets/ahanaf101/bangla-deepfake-dataset
Notebook usage	25,506 files: 12,749 real and 12,757 fake
License	Must be manually confirmed from Kaggle before final submission
Known limits	Possible speaker, recording-domain, and generator overlap

Model	Details
Type	XGBoost binary classifier
Input	227 handcrafted acoustic features
Preprocessing	Mono, 16 kHz, first 5 seconds
Scaler	Saved StandardScaler from notebook
Assets	models/bangla_deepfake_detector.joblib; models/audio_scaler.joblib
Output	Real and fake model probabilities with three-way decision band
Reported result	99.57% accuracy on notebook random held-out split

Intended Uses	Out-of-Scope Uses
Screening, media triage, awareness, research demonstration, human-in-the-loop verification support.	Criminal conviction, automatic censorship, biometric identity verification, proof of speaker identity, unattended high-stakes decisions.

Ethical and Technical Limitations
False positives and false negatives are possible. Performance on unseen TTS systems, unseen speakers, compression, noise, music, overlapping speakers, and phone-quality audio is not established. The random split may overestimate field performance. Uploaded voice data is sensitive and should be handled with consent and care. The tool is AI-assisted screening, not legal or forensic proof.

Recommended next validation: external challenge set, speaker-disjoint split, cross-generator testing, compression/noise robustness table, and consented public examples.